

G. Gaza

time limit per test: 3 seconds
memory limit per test: 1024 megabytes

The Gaza Strip or simply Gaza, is a polity and the smaller of the two Palestinian territories (the other being the West Bank). On the eastern coast of the Mediterranean Sea, Gaza is bordered by Egypt on the southwest and Palestine on the east and north. The Gaza Strip is 41 kilometres (25 miles) long, from 6 to 12 km (3.7 to 7.5 mi) wide, and has a total area of 365 km2 (141 sq mi). With around 2 million Palestinians on approximately 365 km2 (141 sq mi) of land, Gaza has one of the world's highest population densities.

Given an array a of n elements, and q queries. Each query is in one of the following formats:

- 1 i X ($1 \leq i \leq |a|$), ($0 \leq X \leq 10^9$): to change element at index i to the **last bit** of (a_i Xor X), i.e $1\&(a_i \oplus x)$.
- 2 l r ($1 \leq l \leq r \leq |a|$): to calculate decimal representation from l (Most Significant Bit) to r (Least Significant Bit) in the array if possible.

You have to print the output for all queries of the second type if possible, otherwise, print "*Impossible*" without quotations.

If the answer is too huge, output the result $mod\ 10^9 + 9$.

Input

The first line contains a positive integer n ($1 \leq n \leq 10^5$), the number of elements in the array.

The second line contains n values, and the i^{th} element a_i in the array a . ($0 \leq a_i \leq 10^9$)

The third line contains a positive integer q ($1 \leq q \leq 5 \times 10^5$), the number of queries. Followed by q lines containing the queries, one per line. Each query is given in the format described in the problem statement. It is guaranteed that there is at least one query of the second type.

Output

For each second query type, output the decimal representation, or "*Impossible*" if not possible, in a separate line.

Example

input	Copy
5 1 2 3 8 6 8 1 2 5 2 1 3 1 3 5 2 1 3 2 1 5 1 4 5 1 5 9 2 1 5	
output	Copy
Impossible 6 Impossible 27	

Note

It's only eligible to apply this operation on the array's elements of values 0s or 1s.

For the first example, we have our array: [1, 2, 3, 8, 6].

After the first query, it will be [1, 1, 3, 8, 6] as $2 \oplus 5 = 7 = (111)_2$, so the last bit 1 will replace element 2.

After the second query, we can't get the decimal representation because of the element 3.

After the third query, it will be [1, 1, 0, 8, 6] as $3 \oplus 5 = 6 = (110)_2$, so the last bit 0 will replace element 3.

After the fourth query, the output will be $6 = (110)_2$.

...

After applying all queries of the first type, the array will be [1, 1, 0, 1, 1].

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